Architectural Particular Specification

STONE WORKS

SECTION 8.5

Rev	Date	Description
0	14 APR 2022	First Issue
A	13 MAY 2022	Tender Addendum No.2

Main Contract for
Proposed Data Centre Development
at 2-10 Tai Yuen Street, Kwai Chung, N.T.
Andrew Lee King Fun & Associates Architects Limited

SECTION 8.5

STONE WORKS

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. The works shall include, but not limited to, design, testing, supply, fabrication, sealing, transportation, assembly and installation so as to provide a completed stone work package, in full conformity with the requirements and design intent of drawings and this specification.

B. Embedded Concrete Anchors and Inserts

Should cast-in type embed or anchors to be used, all these cast-in fixtures must be furnished and installed by the Main Contractor during construction of the superstructure. The Main Contractor shall be responsible for checking the set out of anchors before concrete pours. Inserts to be embedded in the concrete shall be channel type inserts or approved equal. Attachments and supports embedded into the stone facing shall be stainless steel. All fasteners will be stainless steel.

C. Anchor Bolt

Should post-concreting work anchor bolt type of anchorage system to be used, all anchor bolt design, details, calculation, shop drawings, supply and installation works should be completed by the main contractor. All anchor bolts shall be installed in full accordance to the approved shop drawings, manufacturer's recommendation, statutory requirements. Any testing works required by any statutory authorities shall be deemed to be part of the main contractor's scope. Suitable planning should be allowed for material procurement and testing work to cope with the main contractor's master programme.

D. Flashing Work

An integral continuously watertight aluminium flashing drained to the exterior, shall be provided at each floor level. Flashing shall be provided with expansion splices as required. Zoning and/or end dams shall be provided to isolate and localize water infiltration. Drainage slots or weepholes shall be equipped with filter foam baffles, securely attached to the flashing overlapped and tight fitted to the drainage openings provided.

E. Cladding Coordination

The Main Contractor shall coordinate the sizes and details of stone cladding units.

F. Weatherproof Sealant

1. Stone shall be fixed and sealed weatherproof.
2. Sealant shall be silicone or urethane. Care shall be exercised in selection and application of sealant to avoid atmospheric dirt accumulation, staining of stone or compatibility problems between dissimilar sealants.
3. The waterproofness of the backing wall and structural stability of the stone cladding are the prime responsibility of the main contractor. Any defect or water seepage found within the Warranty Period shall be sealed and made good all at the expense of the main contractor. The backing wall shall be treated with

2 coats water repellent coating by main contractor. This main contractor shall also make good the water repellent coating at fixing points.

G. Cladding Water Runoff

The Main Contractor shall design to collect and drain away all water infiltration through stone and/or supporting walling of the exterior of the facade.

H. Fixing Brackets

Stone cladding shall be furnished and installed complete with all supporting steel, fixing devices, closures, trim and sealant work as required for a complete and watertight installation. Proper corrosive protection coats like galvanisation shall be provided to all steel fixing bracket and accessories.

I. Protective Coating

All stone materials for external use shall be properly pre-treated to prevent water infiltration and to avoid trapping moisture. Stone shall be treated with protective coating and / or sealer as per manufacturer's specification and to architect's satisfaction.

In this regard, all stone / granite floor slabs / pavers / tiles, etc. supplied & installed by the Main Contractor shall be with 6 sides sealer; whereas all stone / granite wall tiles / slabs supplied & installed by the Main Contractor shall be with 6 sides sealer, three(3) coatings shall be applied (i.e. all sides protected with sealer and for the back surface). Technical literature of the sealers / protective coatings and their application method are to be submitted for Architect's acceptance prior to works execution.

Any staining of stone or dirt accumulation due to any defect on protective coating shall be made good or having the stone panels replaced all to the satisfaction of the Architect and at contractor's expense.

J. Design Responsibility

The contract documents will indicate design intent, performance requirements and details for preferred profiles only. The main contractor is responsible for design, development of details, structural calculations, preparation of submission drawings and all associated documents for Buildings Department's approval / consent, statutory tests if any, shop drawings, fabrication, installation, warranties, certifications and related documentation, and testings all to the satisfaction of the Architect / Engineer and the Building Authority.

K. All works shall be in full compliance with all relevant statutory requirements including the Building Ordinance are contract requirements.

L. The Contractor shall propose methods and provide accordingly means to prevent against efflorescence.

1.02 REFERENCES

(NOT USED)

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. General

Prototype portions of each major exterior wall system shall be mocked-up and tested prior to fabrication in order to verify the suitability of the material, structural performance, reaction to thermal cycling, water and air tightness, and resistance to condensation. The testing should be witnessed by representatives from the Client.

All work shall be performed in accordance with the applicable Building Codes, or the requirements of this Specification, whichever are more stringent. If conflict occurs, the contractor shall notify the Architect immediately. Comply with all laws, ordinances, rules and regulations and orders of any public authority having jurisdiction over this part of the works.

Stone flooring and installation system shall conform to, but not limited to BS5385: Part 5 1994.Slip-Resistance Requirements

B. Structural Properties

1. The work shall be designed to withstand the windload as stipulated in the relevant statutory documents but in no case be less than a uniform loading (inward or outward) of 300kg/m^2 for external area.
2. Works for internal areas shall be able to take up the impact loads exerted from the intended use of the areas.
3. The local wind pressures for the design of exterior wall shall be obtained from a wind tunnel test undertaken by a qualified Wind Tunnel Test Laboratory subject to the approval of the Architect/Engineer.
4. Design pressures shall be obtained, but not limited to the following:
   a. Building Enclosures
   b. Ceilings and Soffits(where exposed)
   c. Canopies
   d. Roof Uplift
5. Load Test Assembly Factor

   The entire exterior enclosure will withstand 1.5 times positive and negative loading pressures without assembly performance failure.
6. Stone Modulus-of Rupture Factor (SMRF)

   Stone thickness will be based on a stone modulus-of rupture factor as determined by testing samples of the actual stone proposed for use.
7. Safety Factor (S.F.) Granite

Exposure	Application	S.F.
Exterior	Vertical	2.5
Exterior	Horizontal	5.0

8. Anchorage System Factor

   The exterior enclosure assembly anchorage system to the supporting structure will have a safety factor of 1.5 for both positive and negative loading pressures without assembly failure.
9. Natural Stone Supporting Wall Deflection

   Maximum deflection normal to the wall plane of L/600 at the design wind pressure or a maximum of 6mm in any direction.

C. Waterproofing Against External Weather

1. The work shall be designed to be weatherproof, rigid and stable from such elements as typhoon, rain etc. at such wind load as stipulated under the latest COP on Wind Effect issued by the Building Authority.

2. Exterior Wall Water Penetration

   There will be no appearance of water other than condensation on any interior part of the exterior wall when tested at a pressure of 20% of the design wind pressure or 600Pa, whichever is greater.

3. Exterior Wall Air Infiltration/Exfiltration

   Not to exceed 0.0003 cu.m/s per sq.m on a complete module or bay of the exterior wall when tested at a pressure of 300Pa.

D. Provision for Thermal and Moisture Movement

1. Thermal Movements

   The exterior building enclosure will withstand expansion and contraction forces resulting from an ambient temperature range of 25^{o}C which may result in exterior wall material temperature ranges exceeding 66 ^{o}C.

2. The works shall be so detailed and constructed as to provide for such expansion and/or contraction of the component due to temperature changes varying from seasonal range of 0^{o}C to 40^{o}C and daily range of 10^{o}C and moisture changes without causing harmful cracking, failure of joints, undue stress on fasteners, water leakage or other detrimental effects.

E. Provision for Structural Movement

The works shall be so detailed and constructed, and methods of attachment to structure shall be done so as to provide and allow for the anticipated deflection and movement of the structure without causing damage or having detrimental effects on the works of this section.

F. Variation of Stone Panels

The works shall be so detailed, manufactured and installed to guarantee only the same block/same batch of stone blocks shall be used for the stoneworks of each room/each functional area. Patterns, grains and colour tones of stone slab must be matched naturally and to the satisfaction of the Architect.

G. Polishness

In case polished stonework is carried out, the installed stoneworks shall be finished to have a good quality and durable polishness which shall be accepted at the discretion of the Architect.

H. Integrity

The works carried out shall have no cracks and all stone edge shall be perfectly cut and aligned without any damages or stains.

I. Support

When required, stone support systems, anchors, and accessories shall be manufactured by a company specializing in the design and fabrication of stone approved by the Architect. Provide all fastening devices, support angles, relieving angles, anchors, coping anchors, dowels, cramps, bolts, nuts, shims, expansion shields, flashing, etc., necessary to properly secure stone to the structure as required.

J. Manufacturer's Qualification

1. The Manufacturer has not less than 30 years experiences in the actual production of specified products;
2. No OEM is acceptable;
3. Complies to ISO 9001: 2008 ; & ISO 14001:2004 + Cor 1 : 2009 Quality and Environmental Management System

K. Installer's Qualification

1. Firm experienced in installation or application of systems similar in complexity to those required for this Project;
2. Acceptable to or licensed by manufacturer.

1.05 ENIVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. Submit the following as required in Section 1.5.

B. The contractor shall submit the following information for approval within 4 weeks after the award of main contract.

1. Detail shop drawing showing the elevations, sections, setting out and details with essential dimensions, fixing detail, method of anchorage and fastening, jointing systems, typical joints, back up material, typical corner treatments and sealant, grouting materials.
2. Adhesion Test of the proposed sealant.
3. Structural calculation and drawings prepared and endorsed by a registered structural engineer shall be incorporated as cross reference to the shop drawing submission. All engineering calculations and design shall be based on the mechanical and physical properties of the selected stone. And to be reviewed by independent checker.
4. Programme on delivery and installation.
5. Stone sample of not less than 600 × 600 for each type.
6. Submit 3 nos. stone samples to show full range of exposed colour and texture to be expected upon completion of work.
7. The Contractor is to submit specifications together with any suppliers / manufacturer’s instructions for the handling, storage, installation, protection and maintenance of materials / products specified in this document.
8. Submit manufacturers / suppliers test certificates confirming compliance with the specified requirements.
9. Show stone pattern, type and finish of stone, large size details and sections where stones about other materials, dimensions, thicknesses and specified details.
10. Show plans, sections and large scale details of all stone elements indicating actual location, thickness and setting-out methods.
11. Full set of the photo record for the stone panels before cutting and the proposed stone cutting arrangement.
12. Stone strengthen, water absorption, etc. technical data and job reference.
13. Submit quality control submittals including:-
    a. Statement of qualifications
    b. Statement of compliance with Regulatory Requirements
    c. Field Quality Control
    d. Manufacturer’s field reports

C. For the installation of stone cladding at external wall of building, the contractor should prepare drawings, structural calculations and all other relevant particulars for submission to Buildings Department in soliciting their approval / consent prior to commencement of works on site. Such documents should show details of thickness, strength, durability, fixings and sequence of support. In any circumstances, requirements as laid down in Building Ordinance / Regulations, Code of Practice, and Practice Note for AP/RSE No. APP-16 should be fully compiled with to the satisfaction of Buildings Department and the Architect. Any delay and extra cost incurred as a result of belated and unsatisfactory submission from the contractor will be the sole liability of the contractor, and no claim in whatever means will be entertained.

D. The contractor shall submit product data including the followings:-

1. Detailed specification of construction and fabrication;
2. Manufacturer’s installation instructions;
3. Certified test reports indicating compliance with performance requirements specified herein;

4. All products are compatible to each other in two different products used in same application;

5. Samples of all materials for approval by the Architect before ordering of materials. Sample of min. 300 x 300mm natural stone and of the same thickness as specified on Drawings shall be submitted for approval at least 4 weeks prior to ordering of material. Sample submission shall also state the origin of the quarry of the stone.

E. The Main Contractor shall submit maintenance method recommended by the stone supplier and provide demonstration on the cleaning and maintenance of the stone material. All the maintenance method and maintenance materials proposed shall readily available from the market.

F. The Main Contractor should submit shop drawings / calculation to Architect / structural engineer's approval such that the cast-in anchors could be installed in place in a timely manner to duly match with their own concreting programme. Any abortive / additional works which may be incurred owing to the main contractor's failure to obtain Architect & structural engineer's satisfaction on cast-in anchors' locations will be at their sole risk.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE, AND HANDLING

A. Carefully pack and load finished stone for shipment and take necessary precautions against damage in transit. Do not use material for blocking and packing which would cause staining or discoloration and exercise care to prevent staining during the storage period.

B. Deliver packaged setting materials in the original packages and containers of the manufacturer. Keep aggregate dry and free from soiling.

C. Deliver all materials to site and store in a protected area.

D. Accept packaged stone, materials on site and immediately inspect for damage staining, and discoloration. In the event of materials being damaged, make arrangements to immediately replace with equivalent and matching materials. Handle stone to prevent chipping, breakage, soiling or other damage. Do not use pinch or wrecking bars without first protecting edges of stone with non-staining resilient material. Lift with wide-belt type slings; do not use wire rope or ropes containing tar or other substances which might cause staining.

E. Store stone panels vertically resting weight on panel edge. Prevent straining of stone materials. Do not stack horizontally. Protect stored stone from weather and construction activities, use waterproof, non-staining covers or enclosures which allow air to circulate around stone.

F. Protect mortar materials and stonework accessories from weather, moisture, contamination with foreign materials and in compliance with manufacturers' recommendations and requirements.

G. All stones shall be identified, immediately after fabrication, by marks that clearly indicate the stones location in the building, as indicated the shop drawings.

1.10 MOCK-UPS

A. This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

A. Physical testing will be required to verify that all natural stone cladding materials, stone attachments and supports, bonding materials, etc., meet the requirements noted herein of relevant British Standards.

B. The Contractor shall employ, at its own expense, an independent testing laboratory to perform test specified below and the test shall be performed in the presence of representatives of the Client:

1. Air Infiltration: ASTM E283. conduct test at 300Pa
2. Water Penetration at Static Pressure: ASTM E331. Conduct test at 20% of the design wind pressure loading or 600Pa, whichever is greater
3. Water Penetration at Dynamic Pressure: AAMA 501.1, except conduct test at same loading required for static pressure test
4. Structural Performance: ASTM E330, performance to full design wind pressure and ASTM E330, performance to 2.0 design wind pressure. In addition, test structural performance to failure
5. Cycled Temperature Test
6. Field Pressure Chamber Air and Water Test
7. Adequate in-situ tests including but not limited to tapping tests and pull-out tests shall be preformed to ascertain the bonding strength of the external and internal wall tiles.

C. The contractor shall submit (or for stone cladding works, cause their sub contractors to submit) the laboratory test reports of stone to indicate the compressive and tensile strength, hardness, absorption, imperviousness, resistance to abrasion and resistance to chemicals and pollutants and that the stone complies with design criteria as specified and is satisfactory for the purpose intended for use in Hong Kong and to the satisfactory of the Architect / Engineer and the Building Authority before ordering or delivery in bulk.

D. The performance and design pressures of the steel fasteners, anchorage bolt, fixing lugs, and screws shall be tested and complies with the design criteria as laid down in the specification and the relevant statutes, whichever is the more stringent.

1.12 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. The contractor shall provide a joint warranty with the stone supplier to the Employer for the following:

   1. Guarantee against faulty workmanship and material, loss of mechanical properties, structural failure, non-uniformity of surfaces, installation failure, unsatisfactory workmanship, corrosion and distribution for 10 (ten) years from the certified date shown on the Certificate of Substantial Completion of the Main Contract issued by Architect.

   2. Guarantee that the stone flooring has been installed in accordance with all manufacturer's recommendations and directions. The warranty shall also state that all work is in accordance with the Specification, as amended by any changes thereto authorized by the Architect, free from defects in materials and workmanship for a period of 10 (ten) years from the certified date shown on the Certificate of Substantial Completion of the Main Contract issued by Architect.

C. The warranty is to be submitted in a form to be approved by the Architect. No bulk materials and systems ordering work shall proceed prior to the Architect's confirmation of no adverse comment on the submitted warranty content and format. It is the Main Contractor's sole responsibility to ensure that the required materials or systems can be ordered according their master programme.

D. All warranty is to include for the cost of removal and replacement of all defective work at no extra cost to the Employer and at such times as designated by the Employer.

E. The water-tightness, moisture resistance, thermal stability, and structural stability of the whole stone cladding system are of prime concerns for the works. Any defect or leakage found within the Defects Liability Period shall be sealed and made good at the expense of the contractor.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURES/ SUPPLIERS

A. Subject to compliance with the Project Contract Documents, provide tiles, paves and required accessories from one or more of the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

1. Fortune Building Material (Eng.) Ltd.

   Tel: 2511 0320
   Kitty Chow (6228 3977, 6333 2255)

2. or approved equivalent

2.02 MATERIAL

A. General

1. Stone Thickness

   Minimum stone thickness will be 20 mm. Based on the properties of the stone as determined by the testing laboratory, the location and anticipated loadings, the minimum stone thickness may be increased.

2. Accessories

   a. Trim, base and accessories shall be proved as required for complete installation as specified therein, on drawings and as recommended by the stone manufacturer.

   b. Mortar materials shall be as recommended by the manufacturer for each stone type, size and application. Mortar colour shall be as selected by the Architect.

   c. For coloured pointing mortar, furnish matching stone ground down to meet grading requirements for sand.

   d. Specified and all other necessary materials and products must be compatible with each other and adjacent surfaces and shall cause no adverse reactions including staining, corrosion or any other effects causing breakdown, non-performance or other damage to materials.

   e. All metals in contact with stone shall be stainless steel. Type A4 grade 320 or 316 to B.S. 1449 : 1967.

   f. Embedded anchors in concrete:-

      1) Anchors embedded in concrete shall be prime painted with system acceptable to Architect or hot dip galvanized rolled steel, or hot dip galvanized cold formed steel. To B.S. 729 : 1971.

      2) Strength of embedded anchors shall be developed by integral projections of the steel anchor, or by welded deformed bars or headed studs.

   g. Fasteners requirements listed are applicable to screws, bolts, nuts, washers, rivets and pins.

      1) Fasteners in contact with stone shall be stainless steel type 304.

      2) Provide lock washers, nylon insert nuts or other locking devices at all bolted connections.

      3) Powder actuated fasteners are not acceptable.

      4) Self-drilling, self-threading fasteners are not acceptable.

   h. Provide anchors of proprietary type for each installation method and application required on the project for each installation method and drawings and calculations. For each type use only the best quality anchors which have a proven record of usage within the industry for the intended application, method of support, attachment and resistance to imposed loads. The following is minimum standard for various anchor types:-

1) Kerf clips shall be formed stainless steel; or extruded aluminium with type A41 clear coat anodized finish with a minimum thickness of 0.0175mm (0.7 mil).

2) Pin anchors shall be stainless steel.

3) Anchors shall provide for differential thermal movement of the stone and the support system.

4) The following types of anchors are not acceptable:

    - Wires.
    - Anchors whose only means of attachment to stone or building is a bed of grout or mortar.
    - Anchors which are secured in a hole with lead wool packing.
    - Anchors which rely solely on a composite filler.
    - The colour shall be a custom colour, subject to the sample approval.

i. Slip Pads: Provide Eel slip, nylatron, high impact polystyrene or approved equivalent slip pads between moving parts at all expansion connections. Provide minimum thickness of 1.5mm (0.0625") for nylatron and polystyrene and 3.0mm (0.125") for Eel. Slip. Do not use nylatron or polystyrene in close proximity to a field weld.

j. Where dissimilar metals are encountered, stainless steel 300 series, high polystyrene or nylon separator, minimum 1.5mm (0.0625") thick shall be used.

k. Paint concealed contact surfaces of dissimilar materials, including metal in contact with masonry or concrete work, shall have a heavy coating of dielectric bituminous paint or other means of separation as recommended by manufacturer and acceptable to the Architect.

l. Touch-up welds of hot dip galvanized anchors, anchor inserts, reinforcement and supports with 0.16mm (2.5 mils) minimum thickness of inorganic zinc coating, Dimetcote 9 manufactured by Ameron Coporation or equivalent acceptable zinc rich primer.

m. Structural Shims : At connections subject to thermal movement or other movement, separate all pairs of moving surfaces with friction reducing pads:-

1) Pads shall be minimum 3mm (0.125") thick, shall sufficiently reduce friction to permit movement, shall be resistant to wear and shall be positively retained in position. Open ended slots are not acceptable. Pads shall not be subject to heat damage from welding or cutting, or to excessive pressure from overtightening of bolts.

2) Shims which transfer shear forces, tending to slide one shim against another, shall be steel plates, set in a staggered pattern and fillet welded to each other and to the adjacent steel surfaces.

3) The shims and welds shall be structurally designed to support the applied loads.

n. Stone Shims: Solid lead, non-corrosive, non-staining, malleable.

3. Steel Framing Members

a. All steel framing members shall be galvanized and as follows unless specified otherwise by the Architect / Structural Engineer.

   1) Steel plates, shapes and bars.

   2) Steel tubing: cold formed.

b. Galvanized steelwork. (Hot dip, galvanized min. 65 micron zinc. coating to B.S. 729 : 1971)

   1) Rolled Carbon Steel Shapes : Hot-dip galvanized.

   2) Cold Formed Carbon Steel : Hot-dip galvanized.

c. Stainless steel : Type A4 to B.S. 1449 : 1967 grade 320 or 316

4. Finish and Cutting

Stone shall have the specified finish and cut on all exposed surfaces; concealed surfaces may be sawn.

All granite panels will be finished as follows:-

a. Polished finish.

b. Honed finish.

c. Flamed finish with water jet, re-polish after flaming.
(Only where indicated on the Architect's drawings and where specified by the Architect).

Where flame finish is to be applied it shall be done in a direction parallel to the width, such as the horizontal dimension of the stone. The finish shall be non-directional with no evidence of the linear tracking of the flaming equipment used.

5. Substructure of Stone Flooring

Substructure of stone flooring, if required, shall be constructed of materials and methods as recommended by stone floor manufacturer.

2.03 FABRICATION

A. Fabrication

a. All stonework shall be executed by mechanics skilled in the trade. All stone shall be well-cured and seasoned before cutting. Cut stone units with bed, unless otherwise approved by the Architect.

b. Stone shall be accurately cut to sizes, shapes, profiles and dimensions. There shall be no deviation from jointing.

c. Exposed surfaces and edges of stone units shall be free from cracks, broken corners, chipped arises, scratches or other defects affecting appearance. Patching or filing not permitted.

d. Backs of stone units shall be sawn to true planes, parallel to face plane.

e. Cut stone units full and true on faces, reveals, beds, joint and top to the full dimension required by drawings. All edges shall be straight and true with sharp and true arises. All stones shall fit together accurately.

f. Make faces of stone units in sample plane flush at joints. All finished surfaces shall be true in line and face.

g. Sawn surfaces and edges shall be cleaned of all rusts stains and iron particles.

h. No patching or use of stone with chipped edges or faces shall be permitted.

i. Thickness:

Provide stone of thickness shown on drawings. Saw-cut back surfaces which will be concealed in the finished work. Provide greater stone thickness than shown where thickness shown is insufficient for the sizes or where extend cut-outs shown decreases effective strength of the remaining material, or for proper and sufficient anchorage, suitable and adequate bearing areas or surfaces.

B. Cutting, Drilling and Fitting

a. Provide holes required for anchors, dowels, other devices required to support and/or suspend stone, and to accommodate other items which connect to or penetrate the stone.

b. Include all cutting, drilling and fitting of stonework required to accommodate the work of other trades. In cutting an fitting, carefully cut and grinde edges to a neat tight fit. Do cutting in such a manner so as not to impair strength or appearance of stone. Use physical templates for all cutting and drilling; obtain required templates from proper trades.

C. Sealer

1. Sealer shall be applied to all 6 sides of the natural stone above substrate.

2. Sealer shall be of penetration type that does not change the appearance of the stone.

3. Properties:

All sealer shall be solvent-base impregnator and its special molecular elements shall penetrate into the surface of natural stone protecting them from seepage's and from prejudicial action of external agents and extraneous substance. It shall develop a high protective, water-proofing and anti-stain action. It shall highly resists against penetration of soils, smog, moulds, coffee, dirt etc. it shall not alter the colour or cause any film on the surface of the treated stone. The water-proofing, anti-seepage properties and its strong stain resistance shall limit or eliminate the damage arising from harmful action of oils, grease smog, moulds, coffee, wines, fruit juice, drinks, humidity and perfumes.

4. Installation:

    1) The surface to be treated shall be dry, without dust and free from any kind of greasy substances. It shall be applied by roller, sprayer and brush.

    2) It shall depend on the environment conditions (temperature, air humidity). The solvent shall evaporate in 10-15 minutes. If the absorption is completed, the excess shall be removed by another clean, dry and not dyed cotton rag, some hard and dried residue remained on the surface (always on the polished surface) shall be easily removed by using a small quantity of the sealer.

    3) For best protecting action, further application shall be made after one hour from the previous one. Depending on the porosity of the materials applied, the coverage shall be about 12-15 sq.m./Litre. Complete protection shall be effective after 24 hours.

5. Standards: sealer shall comply with all applicable air quality management regulations including those restricting Volatile Organic compounds (VOC) content to less than 250g/L(208lb/gal)

6. Delivery: Sealer shall be delivered in the original factory, unopened, undamaged packaging bearing identification of the product, manufacturer, batch number, and expiration data, as applicable.

7. Storage: Sealer shall be stored in a location protected from damage, construction activity, and precipitation in strict accordance with the manufacturer's recommendations.

8. Stone Surface Preparation: Surface to receive sealer shall be free of dirt, oil, oil grease, curing compound, surface contamination, and water puddles.

9. Protection:

    1) Precautions shall be taken to avoid damage or contamination of any surfaces near the work zone.

    2) After final application, floor shall be protected for at least 7 days, from contamination and disturbances.

D. Fabrication tolerances for exposed surface of individual pieces of stone are as follows:-

1.	Length and height or width	:	+/- 1.6mm (0.0625")
2.	Depth of a Saw Cut	:	+/- 2mm (0.078")
3.	Depth of a Drilled Hole	:	+/- 3mm (0.12")
4.	Deviation from flat plane in 1.2m (4') any direction	:	+/- 2mm (0.078")
5.	Deviation from nominal thickness	:	+/- 3mm (0.125")

E. Stone Fabrication at Anchorage

Stone for hand setting shall be supported by continuous or intermittent kerf clips engaging in continuous sawn grooves at two opposite edges or at applied linear blocks, if applicable. Linear blocks shall be of the same stone type and thickness as the stone panel.

1. The clip shall be sized to penetrate the sawn groove and overlap the stone on either side of the groove by at least 12mm (0.5") measured vertically.

2. The theoretical overlap, or bite, based on the worst combination of building and wall movements shall not be less than 9.5mm (0.375").

3. Actual overlap as measured in the field shall not be less than 9.5mm (0.375") or a greater dimension required by design criteria, whichever is greater.

PART 3 EXECUTION

3.01 EXAMINATION/INSPECTION

A. General

1. Material used in the fabrication of this work shall be new and of the best grade obtainable of their respective types.

2. Obtain flooring materials from only manufacturers who will, when requested, send a qualified technical representative to the project site to advise the Installer of proper installation procedures.

3. Examination criteria:

   All examinations, selections, and approvals shall be for the purpose of achieving a final appearance of stone with the greatest possible uniformity, and will be based upon the following criteria:

   1) Colour within approved, pre-selected colour range and finishes

   2) Sequence matching of adjacent stone units, as approved by the Engineer

   3) Only one sources of each type of stone shall be used throughout the work. Stone shall match the type, pattern, colour, texture and finish of samples available for inspection in the office of the Client.

   4) Conformance to approved shop drawings and details within specified dimensions and tolerances

   5) Other criteria as may be specified by the Architect/Engineer.

4. The Main Contractor shall be responsible for taking accurate job site measurements for all dimension related to the work.

5. The Main Contractor shall examine all conditions pertaining to the installation of stone flooring and shall notify the Architect/Engineer of any conditions detrimental to the proper or timely completion of the installation.

6. The Main Contractor will not change sources or brands of materials during the course of the works.

7. Whenever manufacturer's recommendations for installation differ from those specified, the Main Contractor shall immediately notify the Architect/Engineer before proceeding with the work.

B. System Design & Integration

Stone system requirements shown on details establish basic dimensions, profiles and sight lines. Within the limitations established by drawings and specifications, the contractor is totally responsible for the design of the entire system.

All designs and resolutions proposed by the main contractor must be shown on the contractor's shop drawings and demonstrated in the mock-up to the approval of the Architect.

C. Colour Variation

The Contractor shall obtain stone from one quarry with consistent colour range and texture for each variety throughout the project.

1. The Employer reserves the right to inspect and reject the selection of quarried blocks from which stone cladding will be cut, and cladding pieces before they are installed. The Contractor shall arrange all transportation and access for the Employer or his representatives (2 persons) to the quarry and places of

fabrication and provide new material to replace rejected material at no additional cost to the Employer.

2. The Employer's review is for colour, finish and appearance match only during samples submission. All other aspects of stone suitability and performance are the responsibility of the Contractor.

D. Inspection

1. All shop and field materials and workmanship shall be subject to inspection by the Architect and Employer at all times. The Contractor shall check the quality, dimension, colour, finish etc. of the granite supplied to site. Any defects/damages so found shall be replaced immediately at the Contractor's expense. The Contractor shall have enough surplus stone uncut and stored in Hong Kong for replacement for the damaged/defective stone. The Contractor shall be responsible for cutting the stone to size and he shall bear full responsibility for any resultant delay so caused.

2. Inspection at the Quarry:

   Quarried blocks shall be made available for inspection by the Architect and the representative from the Client.

3. Inspection at the Fabrication Plant:

   Production units shall be made available for inspection by the Architect and the representative from the Client. The Main Contractor shall, after approval of the final shop drawings, advise the Architect when production has begun and of the earliest possible opportunity to inspect a representative sampling of production work.

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION/APPLICATION

A. All Stone Cladding shall be:

1. Installed according to the approved shop drawings, set plumb, square level at proper elevation and located in proper alignment with all adjoining work,

2. All members shall be securely anchored in accordance with approval details.

B. All metal components of the stone cladding shall be designed, fabricated and installed ensuring that visual flatness is maintained when viewed from any angle.

C. The stone cladding shall be restrained properly from moving away from designed positions.

D. The contractor shall ensure that shop fabricated items can be accessed onto the site.

E. Where the shop fabricated items cannot be accessed due to site conditions, the items shall be returned to the shop for correction.

F. All stone flooring and trim shall be installed using experienced, skilled workmen who are regularly engaged in the engineering, fabrication, manufacture, finishing, installation, and sealing of similar work to the highest standard of workmanship. The finished appearance of the floor shall be free from cracking, chipping, or any other defect due to faulty workmanship.

G. Identical Workmanship

The workmanship and material quality of the stone cladding on all elevations shall be identical.

H. Wet Fixing

Where wet fixing is applied, lay stones on a full even bed of cement mortar. Use low-alkali cement mortar to prevent efflorescence. Cement bedding surface shall be scratched for key.

3.04 CLEANING

A. At such time as may be directed, the contractor shall remove all protection, make good any damage thoroughly clean all surfaces with suitable cleaning agent, make final adjustments to all moulding etc. and hardware and hand over the entire installation leaving all in first class working order to the satisfaction of the Architect.

3.05 PROTECTION

A. All apparatus, materials, components and accessories shall be:

1. Delivered to the site in a new condition,
2. Properly packed and protected against damage due to handling, adverse weather or other circumstances,
3. As far as practicable, they shall be kept in the packing cases or under protective coverings, tapes or coatings until required for use.

B. Any item damaged in transit or on site will be rejected and shall be replaced at no extra cost to the Employer and no item so rejected will be considered as a reason for extension of time.

C. In the case of stone or components and accessories which originate outside Hong Kong:

1. All items of stone, components and accessories shall be adequately and securely packed for safe transportation with due regard to the climatic conditions encountered in transit and on arrival.
2. The Contractor shall, at the time of shipping each consignment of components and accessories, provide to the Architect in triplicate, packing lists and Bills of Lading which shall contain full statements of the packages consigned, with particulars of the dimensions, weights, contents, shipping and approximate value of each package.
3. British Standard 1133 and supplements, or other comparable and acceptable code, shall be used as a guide for the standard of packing and package required. All bright, polished or plated parts shall be treated with suitable rust inhibitor.

D. The contractor shall allow for all to be suitably protected by encasing, covering up, etc., and to submit to the Architect for his approval of all proposed protective measures to all stone components, accessories, etc., as appropriate against damage due to building operations (such as covering up with adhesive tape and plastic sheeting or 3mm timber plank as required by the Architect the components after installation to prevent from damage from building operations above or falling object, etc.), adverse weather or other causes up to the date of issuance of the Certificate of Practical Completion.

END OF SECTION